Deliverable 6: AI Technology Stack Recommendation Report

6.1 Description of materials

ParkChoiceSG analysis artifacts covering anonymous driver recommendations, device-local preferences, parking sessions and favourites, external parking and routing feeds, anonymous error reporting with a separate quota record, and authenticated operator administration.

6.2 Prompt and original AI response

Text prompt

"Based on the updated design materials for ParkChoiceSG (which include anonymous mobile driver features, high-frequency external API polling, and a secure authenticated Operator console for report management and manual overrides), recommend a technology stack appropriate for an entry-level software engineering team. We are familiar with Java, React, and modular backend architectures."

Original AI response

Recommended Technology Stack:

Mobile Client: Flutter (Dart) for cross-platform iOS and Android mobile app deployment with platform-adaptive navigation intents.

Backend Application Server: Spring Boot (Java) or FastAPI (Python) to manage REST endpoints, JWT-based operator authentication, and background async data polling.

Caching & Queue Layer: Redis for in-memory caching of high-frequency LTA DataMall lot readings and token-bucket rate limiting for user report submissions.

Database Store: PostgreSQL to maintain relational integrity between carparks, historical vacancy logs, operator audit trails, and user error reports.

External Interfaces: OneMap SG API (geocoding), Google Maps Distance Matrix (durations), and LTA DataMall (live availability).

6.3 Written opinion on PostgreSQL

PostgreSQL is appropriate for persistent server-side entities because foreign keys and transactions support consistent report resolutions, operator audit records and source data.

Active overrides still require explicit application rules to protect them during refresh; database transactions alone do not implement this behaviour. For this course project, the team can keep driver preferences, sessions and favourites on the device and use PostgreSQL for server data without introducing every optional service in the recommendation.

Fit with the current model

Area	Current interpretation
Driver data	UserPreferences, SavedLocation, SearchHistoryEntry, ParkingSession and Favourite remain device-local.
Anonymous reporting	ReportQuota holds the installation token separately from ErrorReport.
Overrides	Closures exclude carparks; rate corrections replace charging rules and remain protected during refresh.
External services	The original response names candidate providers. Concrete adapters must support the current arrival-time, routing, prediction and source-freshness requirements.